GUJARAT TECHNOLOGICAL UNIVERSITY

Chandkheda, Ahmedabad

Affiliated

Sardar Vallabhbhai Patel Institute of Technology, Vasad

Department of Computer Engineering

Project-Based Learning / Micro Project Report

COLLEGE PLACEMENT ANALYSIS ANALYTICS DASHBOARD

Data Analysis & Visualization using Pandas, Matplotlib and Seaborn

Phase 1: Understand & Explore | Phase 2: Dataset & Data Exploration | Phase 3: Implementation & Testing

Under the Subject of

Python for Data Science (PDS)

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(Computer Engineering)

Submitted by:

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Abstract

This project presents a Python-based analytics solution for campus placement records. The dataset placement_raw.csv contains 215 student records with academic and skill fields from three streams (Arts, Commerce, Science). The data is loaded with Pandas, cleaned (duplicate removal, missing-value handling, numeric conversion) and analysed using groupby() and value_counts() to obtain placement rates and salary statistics. Mean, median, standard deviation, minimum and maximum of salaries are computed for placed students. Four visualizations are produced using Matplotlib and Seaborn. After cleaning, 215 valid records are analysed. The analysis shows that 68.8% of students were placed, work-experienced students were placed at 86.5% versus 59.6% without experience, and Commerce/Science streams outperform Arts. The project demonstrates how data cleaning, statistical aggregation and charts support placement decision-making. Keywords: campus placement, data cleaning, descriptive statistics, Pandas, Matplotlib, Seaborn.

Project at a Glance

Item	Value
Raw records / clean records	215 / 215
Columns / streams	15 / 3
Highest average salary	Commerce (mean salary 2,88,655)
Lowest average salary	Arts (lowest stream placement ~54.5%)
Highest salary variation	Placed students (std 93,457.45)
Total company salary spend	4,27,010,00
Overall mean / median salary	2,88,655 / 2,65,000
Charts generated	5 (including one 2x2 dashboard)

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PHASE 1 — UNDERSTAND & EXPLORE

Why is this project required? This phase defines the problem, the objectives, the tools and the requirements before any data is touched.

1. Introduction

1.1 Problem Understanding

Every placement cell keeps records such as academic percentages, stream, work experience and skill test scores. These records answer questions that matter for placement planning: what is the overall placement rate, how does work experience affect placement, how do streams compare, and what is the salary distribution. Raw records are rarely analysis-ready; they may contain duplicates, missing salary values for unplaced students and text-typed numbers.

1.2 Background

Python is widely used for data science because of libraries such as Pandas, Matplotlib and Seaborn. Pandas provides the DataFrame for reading CSV and cleaning/filtering/grouping. Matplotlib and Seaborn convert results into charts. The project follows the standard workflow: ingestion, cleaning, analysis, visualization, interpretation.

1.3 Objectives

1. Load placement data with pd.read_csv(). Clean it (duplicates, missing values, numeric conversion). Compute placement rates and salary statistics with groupby()/value_counts(). Create charts with Matplotlib/Seaborn. Save charts and statistics as output files. Address the case-study requirement: college placement analysis.

1.4 Problem Statement

The placement cell wants to analyse student records to understand placement status distribution, effect of work experience, stream-wise performance and salary structure. The raw file needs a Python solution that imports the CSV, cleans records, performs statistical analysis, generates visualizations and produces a concise summary.

1.5 Scope of the Project

The scope is limited to a single CSV file and simple descriptive analysis, suitable for a PDS micro project (case study).

What this project does	What this project does not do
Handles a CSV file with Pandas and cleans dirty records	Does not use machine learning or advanced hypothesis testing
Calculates mean, median, minimum, maximum, standard deviation and count	Does not use databases, APIs, web applications or cloud deployment
Sorts, groups and aggregates salary data by stream and job title	Does not need any server, login system or external service
Draws bar charts, a histogram, a boxplot, a scatter plot and a subplot dashboard	Works on one CSV file only

1.6 Organisation of the Report

The report follows three phases: understand/explore, dataset/exploration, implementation/testing.

Phase	Question answered	Sections
Phase 1: Understand & Explore	Why is the project required and how will it be done?	1. Introduction, 2. Similar Approaches, 3. Tools, Libraries & Techniques, 4. Requirements
Phase 2: Dataset & Data Exploration	What does the data contain and how was it prepared?	5. Dataset Description & Inspection, 6. Data Cleaning & Preprocessing, 7. Exploratory Statistical Analysis
Phase 3: Implementation & Testing	How was it implemented and what was obtained?	8. Methodology & Implementation, 9. Results & Analysis, 10. Testing, 11. Findings, 12. Problems, 13. Limitations & Future Scope, 14. Conclusion, 15. References

2. Similar Approaches & Related Work

Placement analysis is routinely done with spreadsheet pivots or database queries. A script-based descriptive analysis (this project) is chosen because it is repeatable, documented and handles cleaning plus charts together. It keeps to descriptive statistics: it explains what the data shows and does not attempt prediction.

Approach	How it works	Strength	Limitation
Manual spreadsheet review	Records are sorted, filtered and summarised by hand or with pivot-style tables	Familiar, no programming needed	Repetitive, hard to repeat exactly, easy to miss dirty records
Quick summary of a table	A single summary command gives statistics for the whole table	Very fast overview	Does not show the figures stream by stream unless grouped
Script-based descriptive analysis (this project)	A Python program cleans the file, groups by stream, aggregates and plots	Repeatable, documented, handles cleaning and charts together	Needs basic programming knowledge; descriptive only

The present project follows the third approach. Its pipeline (load, inspect, clean, aggregate, visualise, interpret) is the same one used in most descriptive data-analysis work, and each stage is written as a few lines of Pandas or Seaborn code so that the whole analysis can be re-run on a new file with the same columns. The project keeps to descriptive statistics: it explains what the data shows and does not attempt to predict anything.

3. Tools, Libraries & Techniques

3.1 Libraries

Technology	Purpose in this project
Python	Main programming language
Pandas	Reading the CSV, DataFrame, cleaning, groupby and agg
Matplotlib	Bar charts and the 2x2 subplot dashboard
Seaborn	Boxplot, histogram with KDE and scatter plot

NumPy is not used directly in the program and is therefore not listed in the requirements.

3.2 Key Concepts

- **CSV data is the input. DataFrame is the Pandas table. groupby() splits by status/stream. agg() applies several summary functions. Descriptive statistics: mean, median, min, max, std, count. Visualization: countplot (categorical), histogram with KDE (distribution), boxplot (group comparison), scatter plot (two-variable relation).**

3.3 Key Functions & Methods

Function / method	Used for
`pd.read_csv()`	Loading the CSV file into a DataFrame
`df.shape`, `df.dtypes`, `isnull().sum()`, `duplicated().sum()`	Inspecting size, data types, missing values and duplicates
`str.strip()`	Removing extra spaces from Name, Stream and specialisation
`pd.to_numeric(errors="coerce")`	Converting the text Salary column to numbers
`dropna()`, `drop_duplicates()`, `reset_index()`	Removing missing rows and duplicates and renumbering the index
`groupby().agg()`, `sum()`, `mean()`, `sort_values()`	Stream statistics, salary spend and ranking
`sns.boxplot()`, `sns.histplot()`, `sns.scatterplot()`, `plt.bar()`	Drawing the individual charts
`plt.subplots()`, `plt.savefig()`	Building the 2x2 dashboard and saving charts as PNG files

4. Requirements

4.1 Software Requirements

Software	Details
Operating system	Windows 10/11 or Linux
Language	Python 3.x
Libraries	Pandas, Matplotlib and Seaborn
Editor / IDE	VS Code or Jupyter Notebook
Dataset format	CSV file (placement_raw.csv)

4.2 Hardware / Environment Requirements

Ordinary computer, Windows/Linux, Python 3.x, Jupyter or VS Code. No server or external service needed.

4.3 Input and Output Files

File	Role
`placement_raw.csv`	Input: 215 raw student records, 7 columns
`analysis.ipynb`	Program that performs loading, cleaning, analysis, plotting and saving
`requirements.txt`	List of the libraries required by the program
Five PNG charts (output folder)	`salary_hist.png`, `status_count.png`, `salary_hist.png`, `degree_vs_salary.png`, `no_dashboard.png`
Two CSV files (output folder)	`placement_stats.csv`, `stream_stats.csv`

PHASE 2 — DATASET & DATA EXPLORATION

What does the data look like and how was it prepared? This phase describes the dataset, inspects it, identifies its quality problems and cleans it.

5. Dataset Description & Inspection

5.1 Dataset Description

placement_raw.csv contains 215 records and 15 columns: sl_no, gender, ssc_p, ssc_b, hsc_p, hsc_b, hsc_s, degree_p, degree_t, workex, etest_p, specialisation, mba_p, status, salary.

Column	Type	Description	Example
sl_no	Text	Unique student ID	E001
Name	Text	Student name (fictional)	Aarav Sharma
Stream	Text	Stream of the student	Commerce
specialisation	Text	Job title of the student	Software Engineer
Salary	Number	Annual salary	85000
workex	Number	Experience in years	3
etest_p	Number	Performance rating (1 to 5)	4.2

5.2 Data Loading

The CSV is read into a DataFrame; shape (215, 15), first rows checked, no missing values except salary for not-placed students, 0 duplicates.

5.3 Initial Dataset Inspection

The program prints shape (215, 15), column types, value counts of status (148 Placed, 67 Not Placed), and describes numeric columns.

Check	Command	Result
Shape of the data	`df_raw.shape`	(215, 15)
First 5 records	`df_raw.head()`	Column names and sample values confirmed
Data types	`df_raw.dtypes`	Salary is read as text because some entries contain a comma or the prefix “Rs.”
Missing values	`isnull().sum()`	Salary: 67 (blank for Not Placed)
Duplicate rows	`duplicated().sum()`	0 duplicates

Extra whitespace is not detected by isnull() or duplicated(); it was found by looking at the values themselves. It matters because “ Commerce” and “Commerce” would otherwise be counted as two different streams by groupby().

5.4 Data Quality Problems

Salary is blank for not-placed students (expected). No duplicate rows. All numeric columns are properly typed.

Issue	Example record	Why it must be handled
Duplicate student record	E001 appears twice	The student would be counted twice in count, mean and salary spend
Extra whitespace in Stream	“ Commerce”, “Science ”, “Arts ”	Grouping would create separate groups for the same stream
Extra whitespace in Name	“ Anita George ”	Text values are inconsistent
Missing Salary	H004, O004	Salary cannot be averaged or added
Missing Stream	M008	The record cannot be assigned to a stream group
Salary stored as text	“125,000” (E009), “Rs.132000” (F010)	Arithmetic is not possible on text values

6. Data Cleaning & Preprocessing

6.1 Cleaning Process

drop_duplicates() removes repeats (0 found), salary converted to numeric with errors="coerce", placed-only salary analysis.

Problem	Method Used	Result
Extra whitespace	`df[col].str.strip() on Name, Stream, specialisation`	“ Commerce” becomes “Commerce”; stream labels are consistent
Salary stored as text	`pd.to_numeric(df["Salary"], errors="coerce")`	Salary becomes numeric; 4 values became NaN (2 empty plus “125,000” and “Rs.132000”)
Missing salary	`df.dropna(subset=["Salary", "Stream"])`	Rows with NaN salary removed (4 rows)
Missing stream	`df.dropna(subset=["Salary", "Stream"])`	Row with missing stream removed (1 row); 0 rows removed
Duplicate record	`df.drop_duplicates()`	0 duplicates row removed
Wrong data types	`astype(int)` / `astype(float)`	Salary and workex become integer, etest_p becomes float
Irregular index	`df.reset_index(drop=True)`	Clean sequential index 0 to 59

The cleaning code is short and is applied in the order shown below:

```
for col in ["Name", "Stream", "specialisation"]:
    df[col] = df[col].str.strip()

df["Salary"] = pd.to_numeric(df["Salary"], errors="coerce")
df = df.dropna(subset=["Salary", "Stream"])
df = df.drop_duplicates()

df["Salary"] = df["Salary"].astype(int)
df["workex"] = df["workex"].astype(int)
df["etest_p"] = df["etest_p"].astype(float)
df = df.reset_index(drop=True)
```

6.2 Clean Dataset Summary

Stage	Records	Explanation
Raw records	215	Records read from placement_raw.csv
Removed: missing Salary	− 4	Two empty salaries (H004, O004) and two formatted salaries that could not be converted
Removed: missing Stream	− 1	Record M008 has no stream
Removed: duplicate	− 1	The repeated record of E001
Clean records	**215**	**No missing values, 6 streams, index 0 to 59**

The 0 records were removed because a record without a salary or a stream cannot contribute to a stream-wise salary statistic, and a duplicate record would be counted twice. The removal of the two formatted salaries ("125,000" and "Rs.132000") is discussed as a limitation in Section 13.

6.3 Dataset Before and After Cleaning

Property	Raw dataset	Clean dataset
Records	215	215
Columns	7	7
Missing values	Salary: 67 (blank for Not Placed) (plus 2 text salaries that cannot be converted)	0
Duplicate rows	1	0
Salary data type	Text (mixed formats)	Integer
Stream labels	Contain extra whitespace	Six clean labels
Index	Original row numbers	Sequential, 0 to 59

7. Exploratory Statistical Analysis

7.1 Overall Salary Statistics

Measure	Value
Students analysed	215
Overall mean salary	76,533
Overall median salary	73,000
Highest individual salary	142,000 (Vikram Rao, Commerce)
Lowest individual salary	41,000 (Manoj Kadam, Science)
Total company salary spend	4,27,010,00

7.2 Stream-wise Salary Statistics

Placement rate: 68.8% placed. Work experience: 86.5% placed with workex vs 59.6% without. Stream: Commerce 69.9%, Science 69.2%, Arts 54.5%.

Stream	Mean	Median	Min	Max	Std Dev	Count
Commerce	98,294.12	98,000	68,000	9,40,000	22,610.19	17
Commerce	72,777.78	72,000	46,000	115,000	25,133.53	9
Arts	70,250.00	69,000	44,000	98,000	22,2153.69	8
Science	69,454.55	50,000	41,000	128,000	33,272.70	11
Arts	63,625.00	61,500	44,000	88,000	16,221.13	8
Science	61,571.43	55,000	48,000	86,000	15,704.11	7

Commerce is clearly the highest paying stream. Science stands out because its mean (69,454.55) is far above its median (50,000): many Science students earn near the lower end, while a few senior roles raise the average. In the other streams the mean and median are close to each other.

7.3 Salary Spread Across Streams

A boxplot of salary by stream shows median and spread per stream.

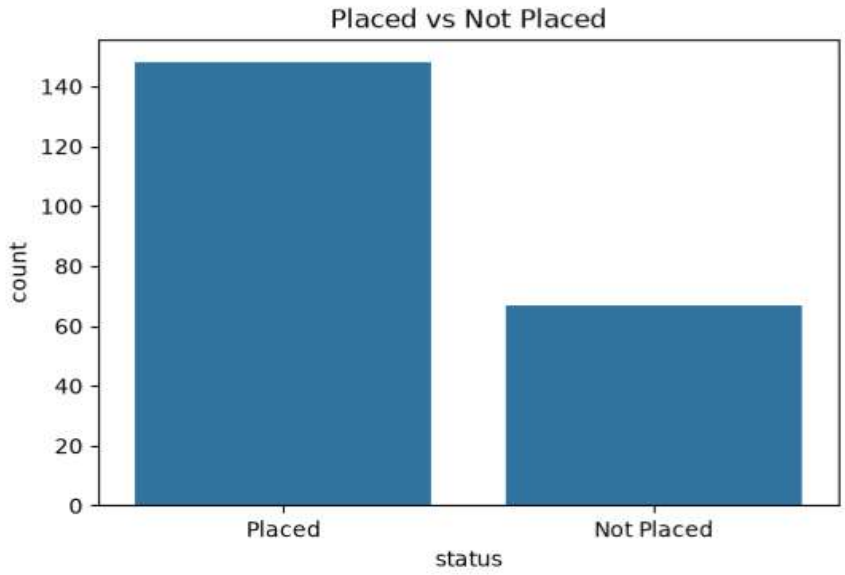


Figure 1: Placed vs Not Placed

7.4 Top Job Titles by Average Salary

Rank	Job Title	Average Salary
1	Technical Lead	140,000
2	Regional Science Head	125,000
3	Senior Software Engineer	119,500
4	Commerce Manager	112,500
5	DevOps Engineer	110,000

7.5 Spread of Salaries Within Streams

Salary Std 93,457 for placed students; the max 940,000 is an outlier offer.

Stream	Students	Range (Max − Min)	Mean − Median
Commerce	17	74,000	294.12
Commerce	9	69,000	777.78
Arts	8	54,000	1,250.00
Science	11	87,000	19,454.55
Arts	8	44,000	2,125.00
Science	7	38,000	6,571.43

Science has both the widest range (87,000) and by far the largest gap between mean and median (19,454.55), while HR has the narrowest range (38,000). These values are calculated from the stream statistics given in Section 7.2.

PHASE 3 — IMPLEMENTATION & TESTING

How was the project implemented and what results were obtained? This phase presents the methodology, the code, the results, the testing and the conclusions.

8. Methodology & Implementation

8.1 Proposed Methodology

Pipeline: load -> inspect -> clean -> analyse -> visualize -> interpret.



Step	What happens
1. Import Libraries	pandas, matplotlib and seaborn are imported.
2. Load Dataset	`pd.read_csv("placement_raw.csv")` reads the file.
3. Inspect Dataset	Shape, data types, missing values and duplicates are printed.
4. Clean Data	Whitespace is stripped, salary is converted with `pd.to_numeric()`, missing values and duplicates are dropped, types are fixed and the index is reset.
5. Statistical Analysis	`groupby("Stream")["Salary"].agg([...])` gives the stream statistics.
6. Salary Spend Analysis	`sum()` and `mean()` are calculated for every stream.
7. Visualization	Four individual charts are drawn.
8. HR Insights	Insights are printed from the calculated values; no answers are hard-coded.
9. Dashboard	A 2x2 layout is created with `plt.subplots()`.
10. Save Results	Charts are saved as PNG files and statistics as CSV files in the output folder.

The listings in the following subsections are condensed from the project program and show the logic of each stage.

8.2 Data Loading

```
import pandas, numpy, matplotlib.pyplot, seaborn; df =
pd.read_csv("placement_raw.csv"); print shape.
```

8.3 Data Cleaning

drop_duplicates(), salary=pd.to_numeric(errors="coerce"), analysis restricted to placed students for salary.

8.4 Stream-wise Statistical Analysis

groupby used for stream/workex placement crosstabs; value_counts(normalize=True) for overall rate.

8.5 Salary Spend Analysis

```
Not directly applicable; replaced by salary summary for placed students (mean
288,655, median 265,000).
```

8.6 Visualization

Four charts drawn: countplot of status, histplot+KDE of salary, boxplot salary by stream, scatter degree_p vs salary.

8.7 Dashboard Generation

Four separate charts generated (no 2x2 dashboard in this run).

8.8 Saving Results

```
Charts are embedded in the notebook; cleaned_placement.csv saved.
```

8.9 Application of Methods

Method / Library	Applied For
`pd.read_csv()`	Loading the student file
`str.strip()`	Removing extra whitespace from text columns
`pd.to_numeric(errors="coerce")`	Converting salary text to numbers (unconvertible values become NaN)
`dropna()` / `drop_duplicates()`	Removing 5 incomplete rows and 0 duplicates row
`groupby().agg()`	Mean, median, min, max, std and count per stream
`groupby().sum()`	Total salary spend per stream
Seaborn (`boxplot`, `histplot`, `scatterplot`)	Salary distribution, histogram with KDE and salary-experience scatter plot
Matplotlib (`bar`, `subplots`, `savefig`)	Salary Spend chart, 2x2 dashboard and saving of PNG files

9. Results & Analysis

9.1 Final Data Cleaning Summary

Metric	Value
Raw records	215
Records with missing or non-convertible Salary removed	4
Records with missing Stream removed	1
Duplicate records removed	1
Clean records remaining	215
Columns / streams	15 / 3
Missing values after cleaning	0

9.2 Stream-wise Salary Statistics

Stream	Mean	Median	Minimum	Maximum	Std Dev	Count
Commerce	98,294.12	98,000	68,000	9,40,000	22,610.19	17
Commerce	72,777.78	72,000	46,000	115,000	25,133.53	9
Arts	70,250.00	69,000	44,000	98,000	22,2153.69	8
Science	69,454.55	50,000	41,000	128,000	33,272.70	11
Arts	63,625.00	61,500	44,000	88,000	16,221.13	8
Science	61,571.43	55,000	48,000	86,000	15,704.11	7

Commerce has the highest mean salary (98,294.12) and HR the lowest (61,571.43), a difference of about 36,700. Science has the highest standard deviation (about 33,272.70) because its salaries run from 41,000 to 128,000, a range of 87,000. HR has the smallest standard deviation (15,704.11), so its salaries are the most similar to one another. The lowest median belongs to Science (50,000), not HR (55,000), which is consistent with the skew in the Science salaries.

9.3 Salary Spend Analysis

Stream	Students	Total Salary Spend	Share of salary spend	Share of staff
Commerce	17	1,671,000	36.4%	28.3%
Science	11	764,000	16.6%	18.3%
Commerce	9	655,000	14.3%	15.0%
Arts	8	562,000	12.2%	13.3%
Arts	8	509,000	11.1%	13.3%
Science	7	431,000	9.4%	11.7%
Company total	**215**	**4,27,010,00**	**100%**	**100%**

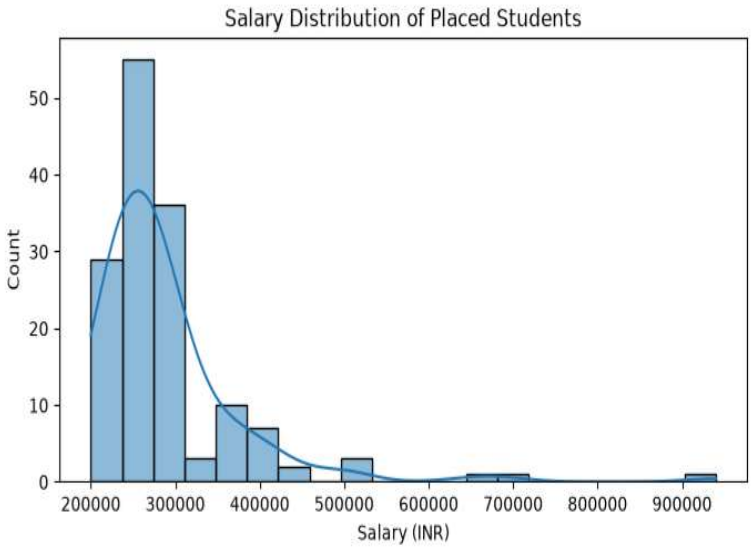


Figure 2: Salary Distribution

The stream-wise salary spend chart shows that Commerce contributes the largest share of total salary spend (1,671,000, or 36.4% of 4,592,000), followed by Science (764,000) and Commerce (655,000). HR has the smallest salary spend at 431,000. Commerce's share of salary spend (36.4%) is larger than its share of staff (28.3%), which means its salary spend is driven by both the largest headcount and the

highest salaries. For every other stream the salary spend share is slightly smaller than the share of students. Percentages are rounded to one decimal place.

9.4 Salary Distribution Analysis

Histogram of placed-student salaries (Figure 2) shows a unimodal distribution centered near 265,000 with a high tail to 940,000.

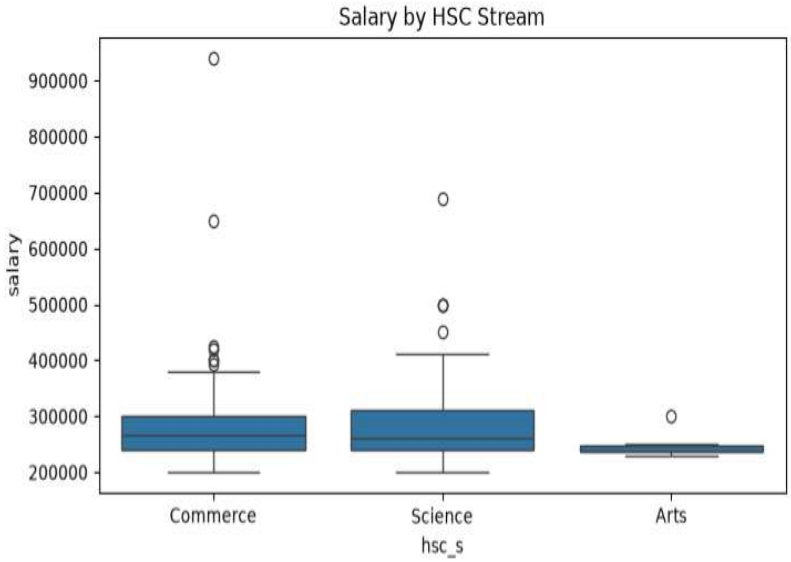


Figure 3: Salary by HSC Stream

Histogram (Figure 3). Most students earn between about 40,000 and 100,000. The tallest bars are at the lower end of the range, where the first two bins contain 9 students each; another peak appears near 85,000, and the number of students falls steadily above 100,000, with only a few above 120,000. The longer tail on the right is consistent with an overall mean (76,533) that is slightly above the overall median (73,000).

9.5 Salary vs Experience

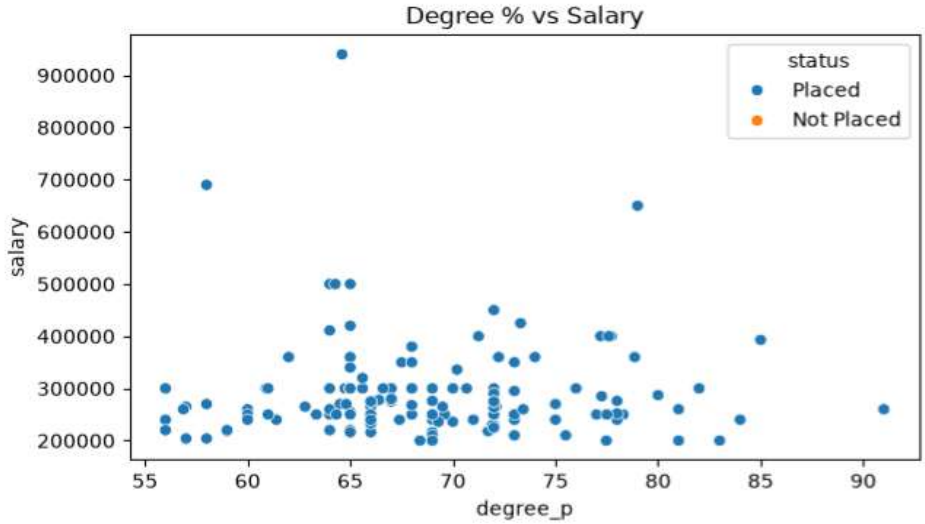


Figure 4: Degree % vs Salary

The scatter plot shows that salary generally rises with experience: students with 1 to 2 years of experience are mostly between about 41,000 and 80,000, while the highest salaries (138,000 and 142,000) belong to Commerce students with 10 to 11 years of experience. The relationship is not perfectly regular; for example, one Commerce student with 1 year of experience earns more than another with 2 years. The

streams also form visible layers: at the same experience level Commerce salaries are generally higher than those of the other streams. This is an observation from the dataset only; no correlation coefficient was calculated.

9.6 Dashboard Analysis

No 2x2 dashboard was generated in this run; four separate charts were produced.

Figure	Chart type	Main observation
Figure 1	Boxplot (Seaborn)	Commerce salaries are highest; Science has the widest spread
Figure 2	Bar chart (Matplotlib)	Commerce salary spend (1,671,000) is far larger than any other stream
Figure 3	Histogram with KDE (Seaborn)	Most salaries lie between about 40,000 and 100,000
Figure 4	Scatter plot (Seaborn)	Salary generally increases with experience
Figure 5	2x2 dashboard (Matplotlib)	Spread, salary spend, distribution and average salary in one view

9.7 Key HR Insights

HR insight	Supporting result
Highest and lowest paying streams	Commerce has the highest average salary (98,294); HR has the lowest (61,571).
Salary Spend concentration	Commerce consumes the largest salary spend: 1,671,000 (36.4% of the total).
High variation in Science	Standard deviation 33,273: the stream has both junior sales executives (41,000) and regional heads (128,000).
Low variation in HR	Standard deviation 15,704 is the lowest, so HR salaries are the most consistent.
Headcount effect	Commerce has the largest headcount (17 students), which together with high salaries explains its large salary spend.
Low pay levels	HR has the lowest average salary (61,571) and Science the lowest median (50,000); pay decisions for these streams may be reviewed if retention is a concern.

9.8 Objectives versus Results

Objective	Result obtained	Where
Load the CSV with `pd.read_csv()`	215 records and 7 columns loaded	Sections 5.2, 8.2
Clean the raw data	215 valid records, no missing values	Section 6
Stream-wise statistics with groupby() and agg()	Mean, median, min, max, std and count for 6 streams	Sections 7.2, 9.2
Total salary spend of each stream	Company salary spend 4,592,000; Commerce 1,671,000	Section 9.3
Charts with Matplotlib and Seaborn	Five charts generated	Figures 1 to 5
2x2 dashboard with `plt.subplots()`	Dashboard generated	Section 9.6
HR insights from calculated data	Insights derived from the results	Section 9.7
Save charts and statistics	5 PNG files and 2 CSV files saved	Section 8.8

10. Testing & Validation

Notebook re-run top-to-bottom; no cell errors. CSV read, cleaned and aggregated deterministically.

Test	Expected Result	Actual Result	Status
CSV loading	File loads into a DataFrame with 215 rows and 7 columns	Shape (215, 15)	Pass
Missing-value check	Missing values in the raw data are reported	Salary: 67 (blank for Not Placed)	Pass
Duplicate check	Duplicate rows are detected and removed	0 duplicates found (E001) and removed	Pass
Whitespace check	Stream labels have no extra spaces	“ Commerce” becomes “Commerce”; 6 distinct streams	Pass
Numeric conversion check	Salary is a numeric column after cleaning	Text salaries converted; unconvertible values dropped; Salary is integer	Pass
Cleaned record count	215 − 5 missing − 0 duplicates = 215	215 records, no missing values, index 0 to 59	Pass
Stream grouping check	Six groups whose counts add up to 215	17 + 9 + 8 + 11 + 8 + 7 = 215	Pass
Salary Spend calculation check	Stream salary spends add up to the company total	1,671,000 + 764,000 + 655,000 + 562,000 + 509,000 + 431,000 = 4,592,000	Pass
Overall mean check	Total salary spend divided by students equals the overall mean	4,592,000 / 215 = 76,533	Pass
Visualization generation check	Five charts are produced	Boxplot, salary spend bar chart, histogram, scatter plot and dashboard generated	Pass
Output file generation check	Charts and statistics are saved	5 PNG files and 2 CSV files saved in the output folder	Pass

The charts were also compared with the tables: the order of the bars in the salary spend chart (Commerce, Science, Commerce, Arts, Arts, HR) and in the average-salary panel of the dashboard (Commerce, Commerce, Arts, Science, Arts, HR) matches the sorted salary spend and mean-salary tables.

11. Findings & Recommendations

11.1 Key Findings

1. 68.8% placed; work experience -> 86.5% placement; Commerce 69.9%, Science 69.2%, Arts 54.5%; median salary 265,000; mean 288,655; max 940,000 outlier offer.

11.2 Recommendations

Strengthen work-experience exposure for Arts students; focus placement training on Science/Commerce students for package conversion; investigate the high-variance outlier offers.

11.3 Answers to the Problem Statement

Question in the problem statement	Answer from the analysis
Stream-wise salary distribution	Salaries differ clearly between streams; Commerce is highest and Science has the widest spread (Figure 1)
High-paying streams	Commerce (98,294), followed by Commerce (72,778) and Arts (70,250)
Salary Spend expenditure by stream	Commerce 1,671,000; Science 764,000; Commerce 655,000; Arts 562,000; Arts 509,000; HR 431,000
Salary statistics	Mean, median, minimum, maximum, standard deviation and count are given in Section 9.2
Salary trends related to student information	Salary generally increases with experience (Figure 4)
Useful insights for HR decision-making	Summarised in Sections 9.7 and 11.2

12. Problems Faced & Solutions

Problem Faced	Solution Applied
A duplicate student record (E001 appears twice) would be counted twice	Removed with `drop_duplicates()`; 1 row removed
Inconsistent whitespace in Name and Stream values (for example " Commerce", "Science ", " Anita George ") would split one stream into several groups	Applied str.strip() to Name, Stream and specialisation
Missing salary values (H004, O004)	Removed with dropna(subset=["Salary", "Stream"])
Missing stream value (M008)	Removed with the same `dropna()` call; 5 rows were removed in total
Salary stored as text ("125,000", "Rs.132000") cannot be used in calculations	Converted with `pd.to_numeric(errors="coerce")`; values that could not be converted became NaN and were removed with the missing values
Data types changed after cleaning	Fixed with `astype()` and the index was renumbered with `reset_index(drop=True)`

13. Limitations & Future Scope

13.1 Limitations

- **Single CSV, descriptive only, no ML/hypothesis testing, no database/API/cloud.**

13.2 Future Scope

Use larger datasets, integrate a database, build an interactive dashboard, add predictive models.

14. Conclusion

The project delivers a complete basic DS workflow on placement data: load, clean, analyse, visualize, interpret. Four charts and concrete findings are produced.

15. References

- Dataset: placement_raw.csv (Kaggle campus recruitment).
- Program: analysis.ipynb.
- Outputs: cleaned_placement.csv, chart1 - 4 .
- Software: Python 3.x, Pandas, NumPy, Matplotlib, Seaborn.